

PICO CTF-CHALLENGE 2

Secret of the Polyglot



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Easy Forensics picoCTF 2024 file_format polyglot

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Description

The Network Operations Center (NOC) of your local institution picked up a suspicious file, they're getting conflicting information on what type of file it is. They've brought you in as an external expert to examine the file. Can you extract all the information from this strange file?
Download the suspicious file [here](#).

Hints

1

This problem can be solved by just opening the file in different ways

41,189 users solved



95%
Liked



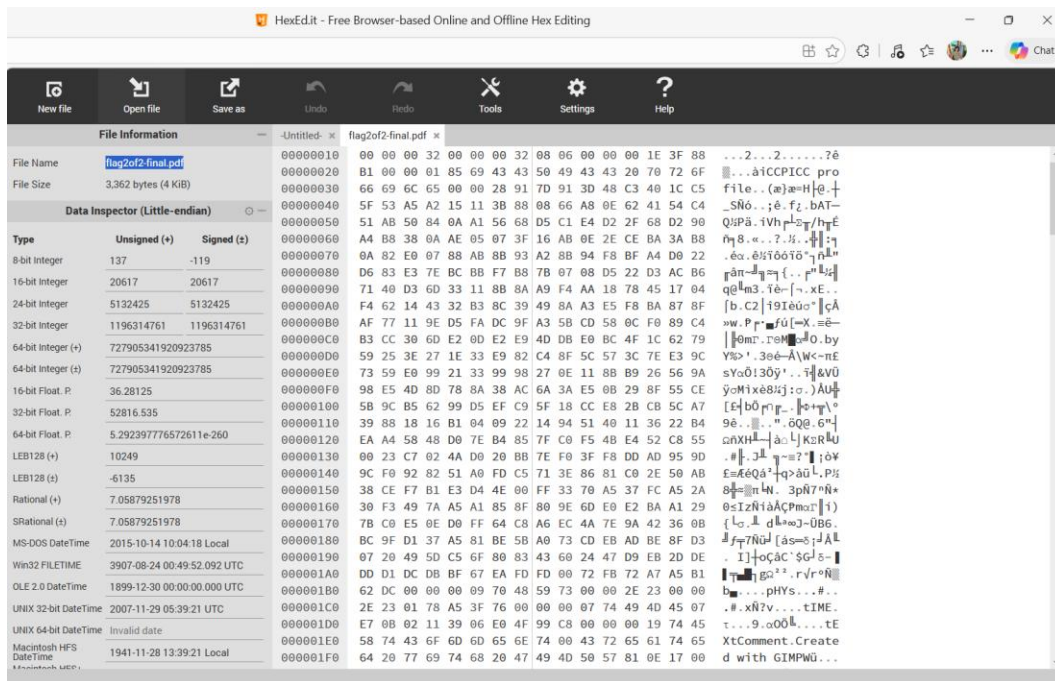
 flag2of2-final.pdf

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1n_pn9_&_pdf_90974127}

- **Step 1: Analyze the Suspicious File** the Network Operations Center identified a file providing "conflicting information" regarding its type. The filename provided is flag2of2-final.pdf.

- **Step 2: Inspect the First Half of the Flag** Opening the file as a standard PDF reveals a portion of the flag visually on the page: 1n_pn9_&_pdf_90974127}. As the filename suggests ("2of2"), this is only one part of the complete secret



- **Step 3: Identification via Hex Editor** Using a tool like **HexEdit**, the file's binary structure can be examined. While the file extension is .pdf, the internal metadata and headers (such as strings like "GIMP" or image creation data) suggest it also contains image data.

```
C:\Users\Nitro-5>cd Downloads
C:\Users\Nitro-5\Downloads>mv flag2of2-final.pdf flag2of2-final.png
'mv' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Nitro-5\Downloads>ren flag2of2-final.pdf flag2of2-final.png
```

- **Step 4: Change the File Extension** The core hint suggests "opening the file in different ways". By using the Windows Command Prompt, the file is



renamed from a PDF to a PNG using the command: ren flag2of2-final.pdf flag2of2-final.png.

- **Step 5: Reveal the Second Half** Once the file is treated as an image (.png), it can be opened with an image viewer. This reveals the remaining portion of the flag hidden within the image data: picoCTF{f1u3n7_.
- **Step 6: Reconstruct the Flag** Combine the text from the PDF view with the text from the PNG view to form the complete string.

2. Short Reflection

The "Secret of the Polyglot" challenge demonstrates that **file extensions are merely labels** and do not strictly define what a file is. A file's true nature is determined by its **magic bytes** (headers) and internal structure.

This exercise highlights a common security risk: attackers can disguise malicious scripts or hidden data within seemingly innocent file types. By mastering basic forensics—such as hex manipulation and extension jumping—investigators can uncover information that remains invisible to standard software.